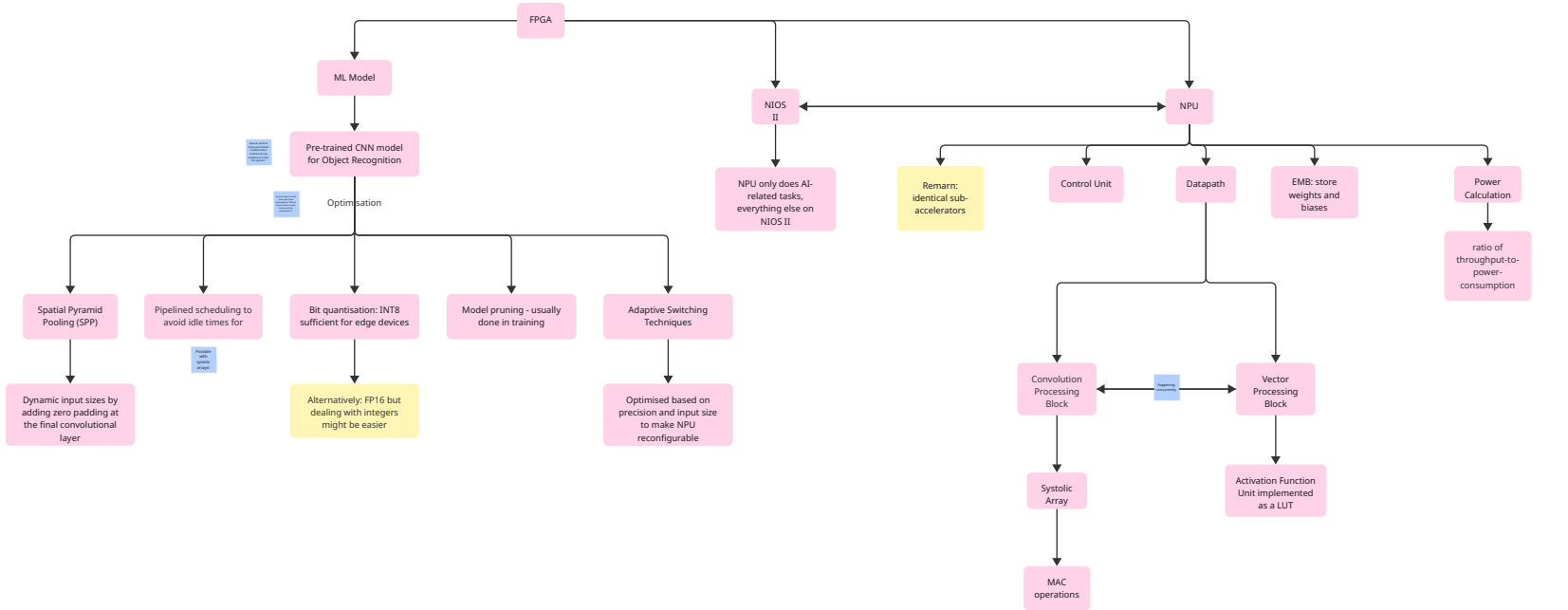
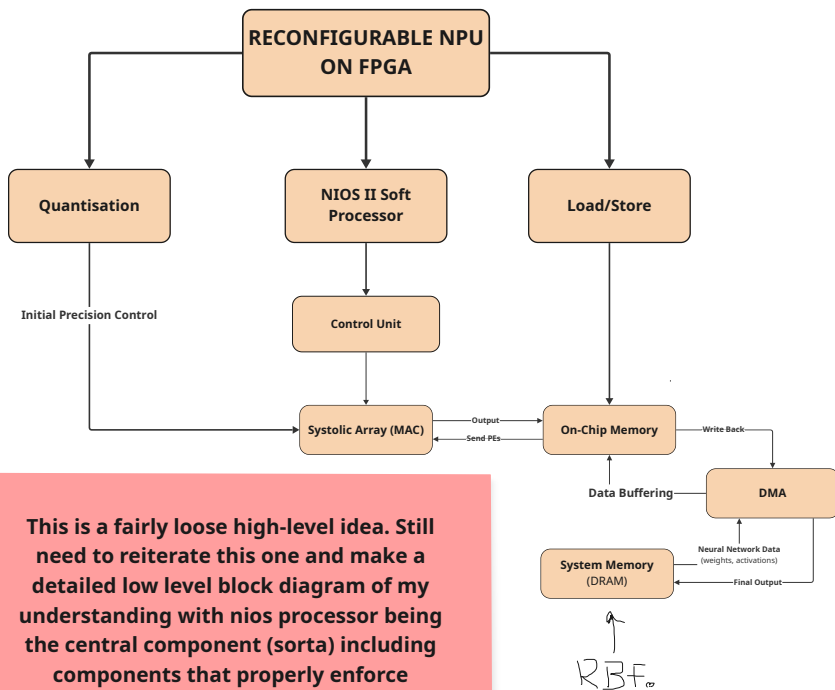






This is a fairly loose high-level idea. Still need to reiterate this one and make a detailed low level block diagram of my understanding with nios processor being the central component (sorta) including components that properly enforce dynamic partial reconfiguration.

NIOS II Soft Processor:

- Acts as a controller on the FPGA, configures the system and basically manages everything. (a "host processor" if you must)

Control Unit:

- FSM to control and manage the execution flow of the NPU operation

Load/Store:

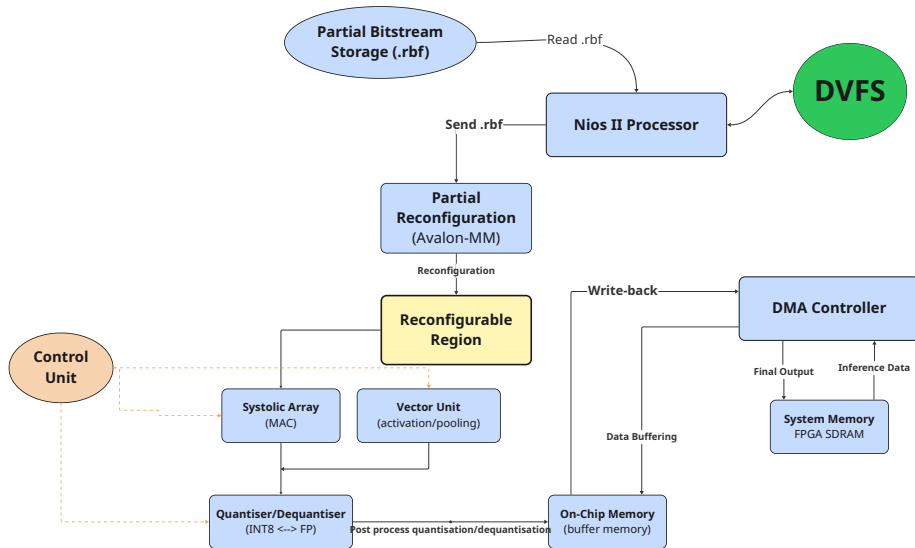
- handles data movement
 - **On-Chip Memory:**
 - Memory local to the NPU used to buffer the data for the MAC component to reduce latency from DRAM.
 - **DMA (Direct Memory Access):**
 - Memory controller that transfers data inbetween the local and external memory
 - **System Memory:**
 - Holds the neural network model and the inference data.

Quantisation:

- Handles data precision management by adjusting the weights and activations to the desired bit width before data enters the MAC array.

Gotta add more shit than just quantisation, but you get the idea.

I thought we were using pre-trained ML?



Other techniques from research to consider utilising:

- **PLPE (posit-based logarithm-domain Processing Elements):** Can be beneficial for reducing hardware complexity and power consumption for multiplication operations to replace multipliers with adders and shifters when needed. Maybe with the DPR.
- **ICRD (Inter-Intra-Channel-Reuse Dataflow):** Reconfiguring buffer sizes depending on different CNN layers.
- **PSCU (Pointed-Stake-Shaped Code Unit)**
Energy saving technique that compresses incoming data or decompresses before computation. Could be an external component near the memory interface

Partial Bitstream Storage (.rbf)

- Stores multiple precompiled hardware configurations for the difference NPU components (e.g MAC unit) with varying precision.

Nios II Processor

- Acts as a central controller of the system. Reads the .rbf files, controls DMA flow, configuring timers/interrupts, manages the inference flow.

Partial Reconfiguration (Avalon-MM)

- Receives partial bitstreams from Nios II and loads them into the FPGA fabric's reconfigurable region. It enables runtime reconfiguration without halting the entire system.

Reconfigurable Region

- Designated Area of the FPGA that can be dynamically reconfigured. Hosts the MAC array.

Control Unit

- FSM that can sequence the operations of the reconfigurable components, such as the MAC array.

Systolic Array (MAC)

- A matrix of PEs used to perform MAC operations, for CNN inference.

Vector Unit

- Handles element-wise operations for activation, pooling. Can even be loaded via DPR when needed.

Quantiser / Dequantiser

- Converts data between low-precision integer (e.g INT8) and FP. Enables mixed-precision inference while conserving memory.

On-Chip Memory (buffer memory)

- Local memory used for storing intermediate data between operations, used as a buffer memory before data is written back to system memory via DMA.

DMA Controller

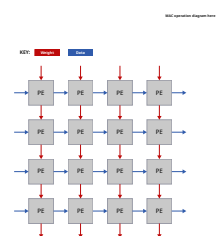
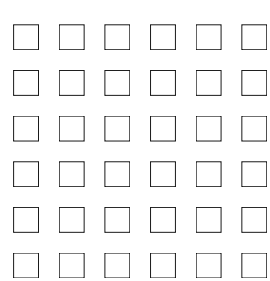
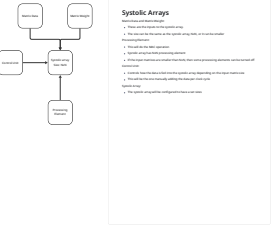
- Independent data transfer between system memory and on-chip memory.

System Memory

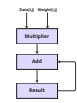
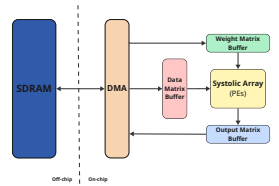
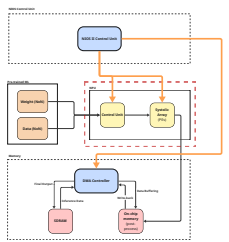
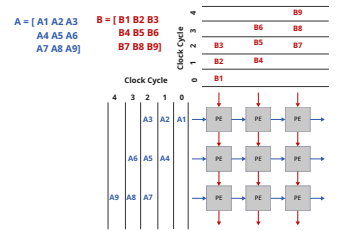
- Stores all neural network data such as weights and outputs of inference results.

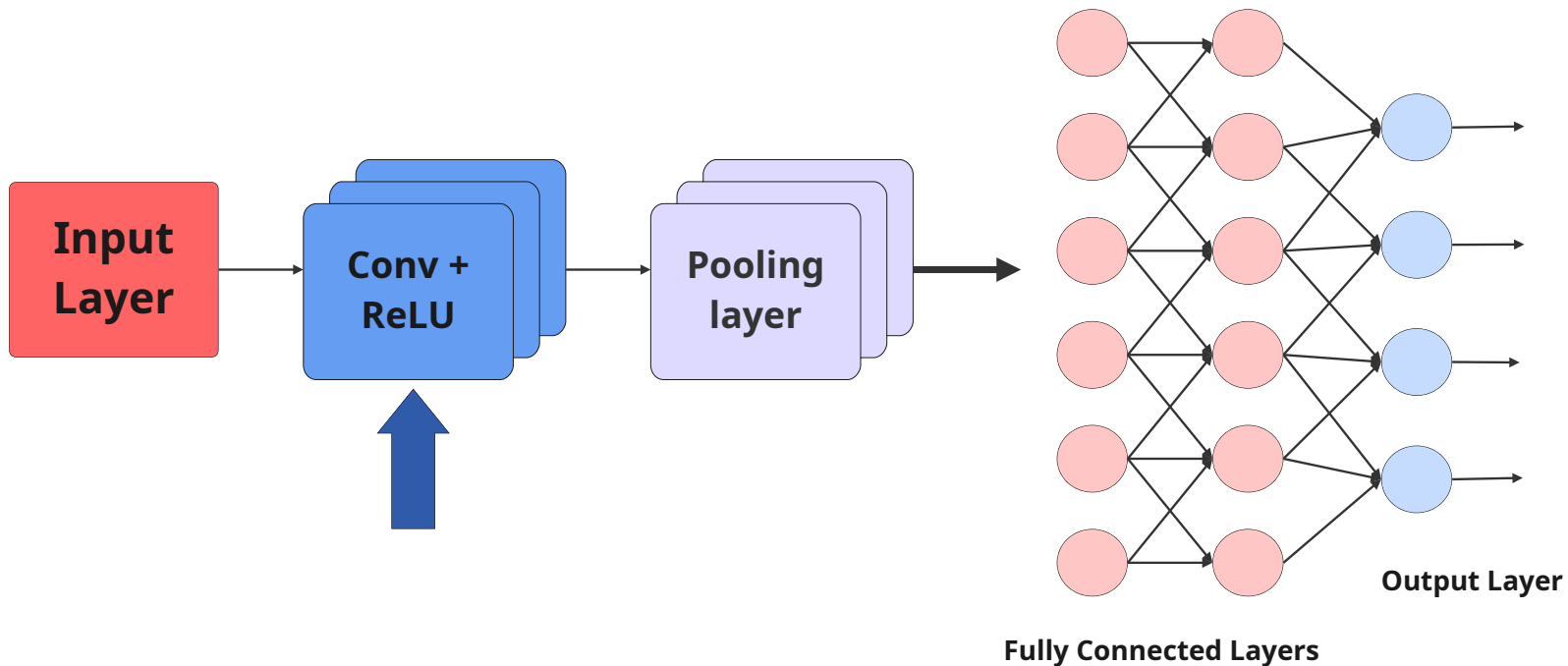
DVFS

- Dynamic Voltage and frequency scaling is a useful power management technique comprising of a dynamic adjustment of the system voltage and frequency based on the computational load. By monitoring the inference workload and system performance.



MAC operation diagram here





3	16	12	14	14	13	9	10
17	4	13	20	27	25	20	1
-42	-48	-32	-36	-6	41	87	121
-18	17	27	44	57	54	39	19
-48	48	46	115	66	17	-23	-44
-20	27	-32	26	-7	4	9	-1
5	-4	-8	24	22	17	6	-13
9	-1	-1	6	-30	-80	-70	-16

Weight Matrix



0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	1
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

Data Matrix



0	0	0	0	0	0	0	0
9	-1	-1	6	-30	-80	-70	-16
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

Result Matrix

	Without Preprocessing and PE gating	With Preprocessing and PE gating
Active Rows	8	1
Active Columns	8	8
K-Value	8	8
Latency (clk cycles)	23	16
Mac Count	64	8

	Without Preprocessing and PE gating	With Preprocessing and PE gating
Active Rows	8	1
Active Columns	8	8
K-Value	8	8
Latency (clk cycles)	23	16
Mac Count	64	8

WEIGHT MATRIX

B = [B1 B2 B3
B4 B5 B6
B7 B8 B9]

Clock Cycle

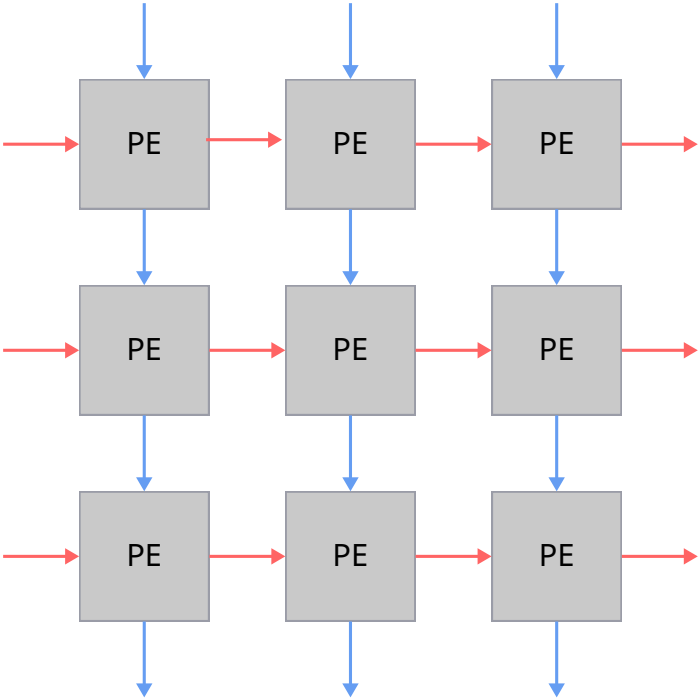
4
3
2
1
0

		B9
B6		B8
B3	B5	B7
B2	B4	
B1		

A = [A1 A2 A3
A4 A5 A6
A7 A8 A9]

DATA MATRIX

4	3	2	1	0
		A3	A2	A1
	A6	A5	A4	
A9	A8	A7		



Data[i,j] Weight[i,j]



Multiplier



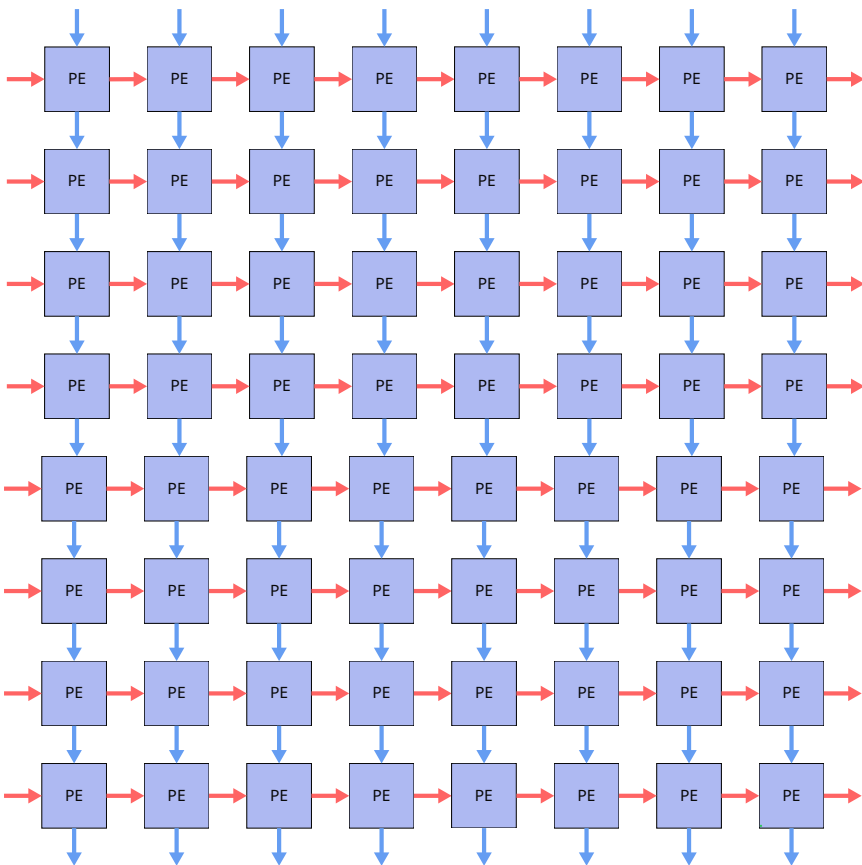
Add



Result



8x8 Baseline Systolic Array Arrangement



KEY:

Weight

Data